Worksheet 1

Assignment 3

Question 1

The dimension of an embedding matrix should have the size of number of word in the training dictionary by the dimension of embedding dimension (dim_vocab*dim_embedding). Each row of the embedding matrix is a vector that uniquely represents a single, non-repetitive word in the training dictionary, and row number equals the number of word in the training dictionary.

Question 2

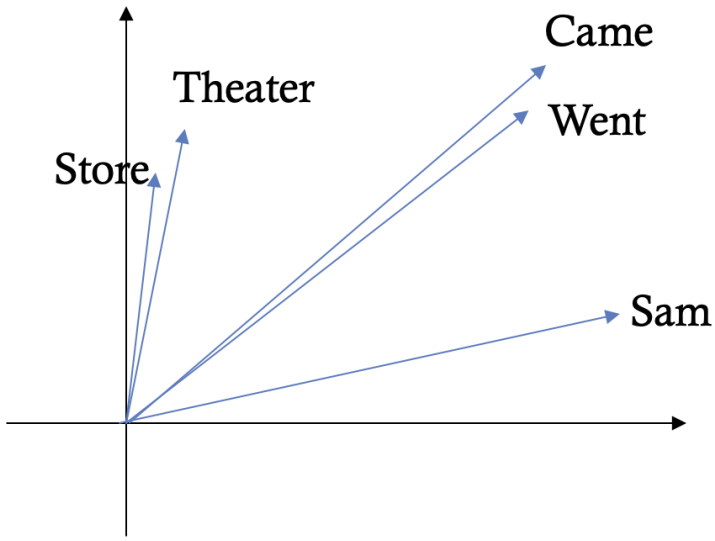


Figure 1: schematic of embedding vector for the chosen words

Question 3

The benefits to use RNN over N-grams is two-fold:
1. The training of RNN is computationally cheaper than that of N-grams models: it avoids to train a huge weight matrix with size of embedding size * (n-1) by vocabulary size.
2. It gives more flexibility to decide how many words are related to the next prediction by passing the hidden state parameters.

Question 4

The LSTM cells are a block of computation that feeds its own output into its input. Specifically, there are two states that will be transfered into the next cell: cell state and hidden state.
In Vanilla RNNs, the input for next RNN cell is the combination of embedding of word x_t and previous state s_{t-1}. The output of that cell is the current state s_t. Such topology only ensures the pass of last step (short-term) memory.
However in LSTM cells, there are three types of modules: forget, remember, and output module. The forget and remember modules ensures the memory from long and short-term can be passed by into the next cell by passing the cell state c_t. Specifically, the forget module can filter out some of the memory by not letting them pass, while the remember module can add some new memory into the slots of current cell state.

Ethical Implications

Question 1

(a) The bias comes from the training data. Since the Amazon's models were trained to vet applicants by observing patterns in resumes over a period, while the most of them were from male's applicants. The mistake was that this model thought the male candidates are more preferable, and penalized resumes that included the words related to women.
(b) Amazon tried to edit the programs to make them neutral to the words related to women and never took the rankings as a sole reference.
The downside is there is no way to guarantee the fairness of these ranking. What's more, there are more fairness issue needs to be addressed, such as diversity, qualifications... And eventually, it is ultimately disbanded as a sole criteria for recruiting new employees. However, on the bright side, it was the first attempt in industry to use AI to recruit new employees, and there are a lot followers.

Question 2

(a) The deployment of this algorithm may bring some positive effects such as:
1. help to create AI writing assistants,
2. more capable dialogue agents,
3. unsupervised translation between languages,
4. better speech recognition systems.

(b) The deployment of this algorithm may bring some negative effects such as:
1. generate misleading news articles,
2. impersonate others online,
3. automate the production of abusive or faked content to post on social media,

4. automate the production of spam/phishing content.
(c) Some malicious actors have begun to target the shared online commons with hateful words or smears to spread fear so that they are afraid to speak.
(d) I agree the decision of OpenAI for non-disclosing their model. Although academically, the creation of such algorithm is a mark of AI of state-of-the-art, using such model could be a bit of dangerous to mislead people and can be used as a political propaganda. I believe the right way to use of such model must be regulated and controlled in some specific area, or just in academic research.